

**INSuREHub
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Final Project Report

Privacy Within Federated Machine Learning Systems

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1 Project Summary

1.1 Top Level Project Description:

This project focuses on exploring privacy-preserving techniques in Federated Learning, a distributed machine learning approach that enables multiple parties to train a shared model without sharing their private data. The project aims to provide a comprehensive understanding of Federated Learning concepts, its working mechanism, and the privacy risks associated with it.

The project will cover various privacy threats and different types of privacy attacks that can occur in Federated Learning, such as data poisoning and backdoor attacks. Additionally, the project will explore different defense mechanisms such as homomorphic encryption and secure multi-party computation, which can be used to mitigate these privacy risks.

Moreover, the project will discuss the concept of differential privacy and its applications in Federated Learning. It will showcase the implementation of label flipping attacks and differential privacy techniques, and their impact on the accuracy of the model.

The project will also cover the concept of dataset security and how it can be ensured in Federated Learning. Finally, the project will provide insights into future work and research directions in privacy-preserving techniques in Federated Learning.

Overall, the project aims to provide a comprehensive understanding of the various privacy-preserving techniques in Federated Learning and their applications in real-world scenarios. The project will draw from a range of sources to provide a detailed and well-researched analysis of the topic.

1.2 Keywords

Federated Learning, Privacy-preserving techniques, Private data, Privacy risks, Privacy threats, Differential privacy, Dataset security, Data poisoning, Backdoor attacks, Defense mechanisms, Homomorphic encryption, Secure multi-party computation, Label flipping attacks, Accuracy, Real-world scenarios, and Research directions.

1.3 Terminology

- Federated Learning: A distributed machine learning approach that enables multiple parties to train a shared model without sharing their private data.
- Privacy-preserving techniques: Techniques that aim to protect the privacy of users' data and prevent unauthorized access to sensitive information.
- Private data: Data that contains sensitive or confidential information that should be kept confidential and not shared publicly.

- Privacy risks: The risks associated with the disclosure or misuse of private data.
- Privacy threats: Various attacks and techniques that aim to compromise the privacy of users' data.
- Differential privacy: A privacy-preserving technique that adds random noise to the data to mask individual user information while preserving aggregate statistics.
- Dataset security: The measures taken to ensure the security and privacy of datasets used in machine learning models.
- Data poisoning: A type of attack in which malicious data is injected into the training dataset to manipulate the model's behavior.
- Backdoor attacks: A type of attack in which a malicious actor inserts a backdoor into the model during training, which can be exploited later to gain unauthorized access.
- Defense mechanisms: Techniques and methods used to protect against privacy risks and attacks.
- Homomorphic encryption: A technique that enables computations to be performed on encrypted data without decrypting it first, thus protecting the privacy of the data.
- Secure multi-party computation: A technique that allows multiple parties to compute a joint function on their private inputs without revealing their inputs to each other.
- Label flipping attacks: A type of attack in which the labels of a subset of the data are flipped to manipulate the model's behavior.
- Accuracy: The measure of how well a machine learning model performs in making predictions.
- Real-world scenarios: Scenarios in which machine learning models are used in practical applications.
- Research directions: The potential areas for future research and development in privacy-preserving techniques in Federated Learning.

1.4 Team Member Responsibilities

Dhyana Sai Reddy Dalli - Conducted extensive research by reviewing and analyzing numerous academic papers, journals, and research articles related to Federated Learning and privacy-preserving techniques, resulting in a comprehensive literature review of existing work and research in Federated Learning that focuses on privacy concerns and their mitigation techniques.

Lakshmi Saieesha Bonthala - Played a key role in the research efforts of the team by conducting an extensive review and analysis of academic papers, journals, and research articles on Federated Learning and privacy-preserving techniques. Contributed to the compilation of a comprehensive literature review that focused on privacy concerns and their mitigation techniques. Additionally, provided valuable

assistance and support to team members in their research work and offered guidance throughout the project.

Neel Pareshbhai Patel - Responsible for designing, coding, and setting up the Federated Learning model for the project. Leveraged state-of-the-art techniques and tools to build a secure and efficient Federated Learning system that enabled multiple parties to train a shared model without sharing their private data. Ensured the smooth functioning and optimization of the system, by implementing rigorous testing and validation procedures. Collaborated closely with other team members to integrate the model into the overall project workflow and to fine-tune its performance.

Nikhil Satish Joshi - Led the research and development efforts of the team, by conducting extensive research on Federated Learning and privacy-preserving techniques in machine learning. Coded and implemented the Federated Learning model, leveraging state-of-the-art tools and techniques to ensure its secure and efficient functioning. Managed the team effectively, ensuring that all members were on track and meeting their targets. Worked closely with the technical director and other stakeholders to ensure that the project was progressing according to plan and that all technical requirements were being met. Acted as a mentor and guide to team members, providing them with support and guidance to ensure that they were functioning at their best

Shrey Saini - Implemented the Federated Learning model and conducted extensive testing using Google Colab. Analyzed the results and generated charts to gain insights into the model's performance and identify areas for improvement. Additionally, conducted research on relevant techniques and methods, staying up to date on the latest developments in Federated Learning and related areas. Used his knowledge to inform about his work on the project, ensuring that it was grounded in the latest advances and best practices in the field. Overall, his contributions were essential to the successful implementation and optimization of the Federated Learning model and the project.

Srinivas Panchangam - Significantly contributed to the research efforts of the team by conducting a thorough and in-depth analysis of academic papers, journals, and research articles on Federated Learning and various privacy-preserving techniques. Played an instrumental role in the compilation of a comprehensive literature review, which focused on the mitigation of privacy concerns. Provided valuable assistance and support to team members in their research work and offered guidance throughout the project, resulting in the successful completion of the project goals.

1.5 Estimated Project Budget

Our team firmly believes that the project can be accomplished using resources that are already available at the University of Texas at Dallas. We have access to a plethora of subscription-based research platforms at no cost.

1.6 Summary of Key Deliverables

Description	Completion Date
Project Proposal	February 6
Phase 1	March 10
MidTerm Presentation	March 10
Phase Two	March 11
Phase Three	April 7
Final Presentation and Report	May 5

2 Motivation

The motivation behind the project is to address the growing concerns over privacy in machine learning, particularly in the context of federated learning. Federated Learning is a promising approach for collaborative machine learning where parties train a shared model without sharing their private data. However, there are still privacy risks associated with federated learning that need to be addressed.

Privacy threats in federated learning arise due to the sharing of model updates, which contain sensitive information about the training data. Attackers can exploit this information to perform various privacy attacks, including membership inference, model inversion, and model extraction attacks. These attacks can lead to the leakage of sensitive information, such as personal identification data, medical records, and financial information.

To mitigate these privacy risks, various techniques have been proposed, including differential privacy, dataset security, data poisoning, backdoor attacks, and defenses like homomorphic encryption and secure multi-party computation. However, the effectiveness of these techniques in federated learning is not well understood, and their impact on the accuracy of the trained model needs to be evaluated.

Therefore, the main motivation behind the project is to provide a comprehensive understanding of privacy-preserving techniques in federated learning, their strengths, weaknesses, and their impact on the accuracy of the model. The project aims to showcase the implementation of label flipping attacks and differential privacy techniques and their impact on accuracy. Additionally, the project intends to discuss future work and research directions to provide practical guidance to researchers and practitioners interested in this field.

3 Literature Review

3.1 APPFL: Open-Source Software Framework for Privacy-Preserving Federated Learning

Federated learning has emerged as a promising approach to training machine learning models on decentralized data. It allows models to be trained at different locations, avoiding the need for data to be transferred to a central location, which is particularly advantageous in domains where data cannot be freely shared or stored centrally due to policy regulations. This has led to a surge in research efforts, resulting in the development of numerous federated learning frameworks. In this context, the Argonne Privacy-Preserving Federated Learning (APPFL) framework has been introduced, which provides users with the ability to leverage existing privacy-preserving algorithms, implement new ones, and simulate and deploy various federated learning algorithms with privacy-preserving techniques. The framework is modular, allowing users to customize components for algorithms, privacy, communication protocols, neural network models, and user data. Additionally, the work presents a new communication-efficient algorithm based on an inexact alternating direction method of multipliers that significantly reduces communication between the server and clients compared to the current state-of-the-art. Experimental results demonstrate the computational capabilities of APPFL and its scalability, including differentially private federated learning on various datasets and algorithms across different computing environments.

3.2 A Taxonomy of Attacks on Federated Learning

The paper titled "A Taxonomy of Attacks on Federated Learning" published by IEEE provides a comprehensive taxonomy of various attacks on the federated learning (FL) framework. The authors discuss different types of privacy attacks and their characteristics in detail. They categorize these attacks into five groups: model poisoning attacks, model extraction attacks, inference attacks, communication attacks, and backdoor attacks. The paper also provides examples of these attacks and the potential impact on FL systems. The authors conclude with suggestions for future research directions to improve the security of FL systems. This paper provides a valuable contribution to the literature on privacy and security in FL.

3.3 Dataset Security for Machine Learning: Data Poisoning, Backdoor Attacks, and Defenses

The paper "Dataset Security for Machine Learning: Data Poisoning, Backdoor Attacks, and Defenses" discusses the security threats faced by machine learning models due to malicious data manipulation. Specifically, the paper covers two types of attacks: data poisoning and backdoor attacks, and presents various defense mechanisms against them. The paper also provides experimental results to demonstrate the effectiveness of these defense mechanisms. Overall, the paper aims to provide insights into the importance of dataset security in machine learning

and presents solutions to enhance the robustness of machine learning models against such attacks.

3.4 Defending against Label-Flipping Attacks in Federated Learning Systems with UMAP

This paper proposes a novel defense mechanism against label-flipping attacks in Federated Learning systems using UMAP (Uniform Manifold Approximation and Projection). The authors demonstrate that UMAP can be used to identify and mitigate label-flipping attacks by projecting the dataset onto a lower-dimensional manifold and then retraining the model. Experimental results show that their proposed approach achieves higher accuracy and better resistance against label-flipping attacks compared to existing methods. The paper provides a comprehensive understanding of label-flipping attacks in Federated Learning and offers insights into potential defense mechanisms.

3.5 Privacy-Preserving Aggregation in Federated Learning: A Survey Ziyao Liu, Jiale Guo, Wenzhuo Yang, Jiani Fan, Kwok-Yan Lam, Senior Member, IEEE, and Jun Zhao, Member, IEEE

The paper presents a survey on privacy-preserving aggregation (PPA) techniques used in federated learning (FL). The authors first introduce the concept of FL and PPA, and then provide an overview of different PPA methods including secure aggregation, differential privacy, homomorphic encryption, and multi-party computation. They also discuss the advantages and limitations of each method and compare them in terms of performance, security, and privacy. The authors then review recent research efforts in enhancing the privacy and efficiency of PPA techniques and provide future research directions. Overall, the paper provides a comprehensive overview of PPA techniques in FL and can serve as a useful reference for researchers and practitioners working in this area.

4 Specific Aims

The aim of this project is to provide a comprehensive survey of privacy-preserving techniques within federated learning models. The project starts by describing the fundamental concepts of federated learning systems, including their advantages, disadvantages, and potential attacks. The team then implements a label flipping attack and analyzes the impact of applying differential privacy techniques to add noise to the data. The findings are presented through graphical analysis of model accuracy across different noise levels and under targeted label flipping attacks.

As part of future work, the project presents two potential areas for improvement in privacy-preserving techniques within federated learning models. First, the team discusses the use of homomorphic encryption as a potential approach to improve the privacy of federated learning models. Second, they suggest exploring the use of secure multi-party computation to enhance both privacy and efficiency of the models. Additionally, the project highlights the potential benefits of incorporating UMAP and weak encryption methods to further improve the privacy of federated learning models.

Overall, the project aims to provide valuable insights into the challenges and opportunities associated with privacy preservation in federated learning models and suggest promising directions for future research. The team hopes that their work will contribute to the development of more effective and secure privacy-preserving techniques in the field of federated learning.

5 Research Approach

5.1 Federated Learning SetUp

Framework Used:

The framework used for this project is TensorFlow Federated, which is designed to enable Federated Learning. TensorFlow Federated (TFF) is an open-source framework for machine learning and other computations on decentralized data.

TensorFlow Federated (TFF) is an open-source framework developed by Google for building machine learning models on decentralized data. It enables the development of machine learning models that can be trained on data that is distributed across multiple devices or clients, while still preserving the privacy of that data. TFF provides a programming interface for building distributed machine learning models and includes support for several different federated learning algorithms, including Federated Averaging, Federated SGD, and Federated Learning with Differential Privacy.

DataSet Details:

The dataset used for this project is the EMNIST dataset, which is a collection of handwritten digit images from 0-9 classes. Specifically, the Federated EMNIST dataset from the `tff.simulation.datasets` package was utilized. The objective of this project was to perform image classification within the Federated Learning framework.

The MNIST dataset is a widely used benchmark dataset in machine learning and computer vision and is considered a classic dataset for image classification. It consists of a set of 70,000 images of handwritten digits, each image being a grayscale 28x28 pixel image. The digits range from 0 to 9, and the dataset is evenly split between training (60,000 images) and testing (10,000 images) sets.

In our experiments, the EMNIST data set was used. It is often used to train Federated ML algorithms. The 10 classes represent 10 digits (0-9). EMNIST is a data set comprising a training set of 60,000 items and a testing set of 10,000 items. For comparison with the same experimental setup, we used the same network architecture of 2 layers with 200, 10 nodes respectively with batch normalization.

Research Approach:

Federated Averaging: Federated Averaging is a method used in Federated Learning, where each device or node trains a local machine learning model on its own data without sharing it with the central server. After each round of local training, the device sends only the updated model parameters to the central server. The central server then aggregates these parameters from all the devices and computes the average. This averaged model is then sent back to all the devices for further

training. This process is repeated iteratively until the model converges to an optimal solution. The main advantage of this approach is that it preserves the privacy of individual devices since their data is not shared with the central server. Instead, only the model parameters are shared.

Federated Learning with Differential Privacy: Federated Learning with Differential Privacy is a privacy-enhancing technique used in federated learning. Differential privacy adds a controlled amount of random noise to the data so that the data cannot be used to re-identify an individual. In Federated Learning, the addition of noise to the model updates before they are shared with the central server helps to protect the privacy of individual devices.

Adding noise to the model updates is done by first computing the gradient of the local model parameters, then adding random noise to the gradient before sharing it with the central server. The central server aggregates the noisy gradients from all the devices and uses them to update the global model parameters. By adding noise to the model updates, the privacy of the data is protected while still allowing for meaningful aggregation of model updates.

The level of privacy protection provided by differential privacy can be controlled by adjusting the amount of noise added to the data. However, adding too much noise can result in a decrease in the accuracy of the model. Therefore, a balance must be struck between privacy protection and model accuracy. Differential privacy is a technique used to enhance the privacy of data. In Federated Learning, differential privacy can be used to add noise to the model updates before they are shared with the central server. This helps to protect the privacy of individual devices while still allowing for meaningful aggregation of model updates.

5.2 Label Flipping Process

Label flipping attacks are a type of data poisoning attack in which the labels of the training data are flipped or modified, causing the model's classification performance to degrade. This attack can be used to reduce the accuracy of the global model in federated learning (FL) even if only a small fraction of participants are under the control of an adversary. To test the vulnerability of the EMNIST dataset to label-flipping attacks, the researchers implemented targeted data poisoning in the FL for label flipping.

We implemented data poisoning in the FL for label flipping. We implemented targeted data poisoning in the FL for label flipping. Specifically, we flipped a label l to l' , where $l \in L$ and $l' \in L$, $l' \neq l$ where L is the number of classes in the classification problem. In this attack, we assigned the number of participants N of which $m\%$ were malicious clients. We randomly selected $N * m$ from participants p where the rest were honest participants at the beginning of each experiment. To improve the precision of the experiment, we dealt with the consequences of the random selection of malicious participants. Each experiment was repeated 5 times, and the average outcome was used as the result.

The rate of malicious client presence (m) used ranged from 2% to 50%. As per the outcomes, the test accuracy of the global model declines as m increases. Additionally, even with a slight increase in m , when compared to a non-poisoned model, the global-model test accuracy decreases, but in this situation, the source-class recall of the model shows a much more significant drop. The global model test accuracy for emnist falls to 57% in the poisoned situation when m is 40%, compared with 78.3% in the non-poisoned model case.

This demonstrates that an adversary can still reduce the accuracy of the global model even if they shift only a small fraction of the participants. As a result, an FL system can be dramatically affected by even a small number of participants under the control of an adversary. The EMNIST data set is vulnerable to label-poisoning attacks.

Defense Mechanism:

Federated learning models are susceptible to poisoning attacks that manipulate the training data to affect the model's behavior. Various defense mechanisms can be employed at different stages of the machine learning pipeline to detect, repair, or prevent such attacks.

The first category of defenses aims to identify the existence of poisoning attacks. One approach is to analyze the training data for anomalies, such as outliers or rare instances that may be indicative of poisoning. Another approach is to monitor the

behavior of the trained model to detect any abnormal or unexpected outputs that may be indicative of a backdoor attack.

The second category of defenses focuses on repairing the poisoned model. This involves removing the backdoor behavior from the model, either by retraining the model on clean data or by removing the poisoned data from the training set.

The third category of defenses comprises robust training approaches that are designed to prevent poisoning attacks from taking effect. These approaches can include using data sanitization techniques, such as outlier detection and removal, or using machine learning algorithms that are inherently resistant to poisoning attacks, such as ensemble models.

Finally, defense mechanisms for federated learning include strategies for detecting and mitigating poisoning attacks in a distributed learning environment, where data is scattered across multiple devices and is not centrally stored. These can include methods for detecting malicious clients and for aggregating model updates in a secure and privacy-preserving manner.

5.3 Dataset Security

The goal of this topic is to catalog and systematize vulnerabilities in the dataset creation process, and review how these weaknesses lead to exploitation of machine learning systems.

The creation of open-world datasets by harvesting information from unverified sources on the web can be exploited in various ways. Outsiders can manipulate datasets by inserting corrupted data either passively or actively. Adversaries can also inject data into systems that rely on federated learning, where models are trained on a network of edge devices that communicate with a central server. Therefore, data creators must take measures to prevent such dataset manipulation and ensure data privacy.

There are 3 possible dataset security issues.

1. **Training-only attacks:** which involve manipulating training data and/or labels and do not require access to test-time inputs after the system is deployed. These attacks can be categorized by the optimization formulation or heuristic used to craft the attack and whether they target a from-scratch training process or a transfer learning process that fine-tunes a pre-trained model. These attacks are particularly relevant in scenarios where training data is collected from potentially compromised online sources, such as social media profiles.
2. **Attacks on both training and testing:** These threats are often referred to as “backdoor attacks” or “trojans.” They embed an exploit at train time that is subsequently invoked by the presence of a “trigger” at test time. These attacks can be further subdivided into model-agnostic attacks and model-specific attacks that exploit a particular neural network architecture. Additional categories of attacks exist that exploit special properties of the transfer learning and federated learning settings.
3. **Defenses against dataset tampering:** Defense methods can either detect when poisoning has taken place or produce an unaffected model using a training process that resists poisons. Detection methods include those that identify corrupted training instances, in addition to methods for flagging corrupted models after they have been trained. Training-based defenses may avoid the consequences of poisoning altogether using robust training routines, or else perform post-hoc correction of a corrupted model to remove the effects of poisoning.

Applications of Data Poisoning:

1. **Federated learning:** The distributed nature of federated learning raises a number of unique issues in dataset security. Because model updates are harvested from a diffuse network of untrusted users, very strong adversarial threat models are realistic. For example, it is reasonable to assume that an adversary has complete control over input data to the system, labels on training data, and model updates broadcast to the central server. At the same time, federated learning adversaries are weak in that they only have access to a small slice of the training data – the entire training system comprises many users that sample from diverse distributions unknown to the attacker.
2. **Reinforcement learning:** Reinforcement learning algorithms are also susceptible to poisoning attacks.
3. **Recommendation systems:** One well-studied application of data poisoning is recommendation systems. the attacker modifies training data either to degrade accuracy overall or to promote a target item at test time.
4. **Differential privacy:** Differentially private (DP) training is a framework for learning from data without relying heavily on or memorizing individual data samples. While originally developed to preserve user privacy, DP training also conveys a degree of resistance to data poisoning since modifications to a small number of samples can have only a limited impact on the resulting model. At the same time, differential privacy can also enable poisoning, as attackers can mask their behavior by manipulating data and model updates before DP data aggregation is applied to harvest data on a privacy-preserving central server.

5.4 Defense Implementation - 1

Our research work focused on the use of dimensionality reduction techniques, specifically PCA and UMAP, as defense mechanisms against data poisoning attacks in federated learning systems. Data poisoning attacks involve adversaries injecting malicious data into a training dataset to manipulate the resulting model's behavior.

In our study, we evaluated the effectiveness of PCA and UMAP in detecting and mitigating data poisoning attacks in federated learning systems. We found that UMAP outperformed PCA in terms of detecting subtle relationships between data points and preserving the geometry of the data. This is particularly important in detecting data poisoning attacks that involve changing the distribution of the data.

However, we also noted that the effectiveness of these techniques depends on the specific attack scenario and the characteristics of the data being analyzed. Therefore, it is essential to carefully consider which technique is most appropriate for a given situation.

Our findings suggest that using UMAP as a defense mechanism against data poisoning attacks in federated learning systems can be highly effective. However, further research is needed to explore the effectiveness of other dimensionality reduction techniques and to evaluate their performance in different attack scenarios.

5.5 Defense Implantation - 2

Federated learning is a distributed machine learning approach where multiple devices collaboratively train a model without centralizing the training data on a server. While this approach offers numerous advantages such as preserving data privacy, it is vulnerable to privacy leakage attacks. To address this issue, the paper proposes a privacy-preserving solution that uses a combination of differential privacy and weak encryption.

Differential privacy is a technique that adds random noise to data to prevent adversaries from inferring private information. The proposed solution perturbs the privacy features using differential privacy to ensure that malicious clients cannot obtain private information during the training process. Weak encryption is used to process the update parameters, ensuring that communication and aggregation of parameters do not reveal any sensitive information. The global parameters are decrypted with all clients, and the updated parameters are re-encrypted and sent back to the server after completing the training. This approach effectively reduces the risk of internal and external attacks while maintaining the accuracy of the training task.

The proposed solution has advantages over other privacy-preserving methods such as Secure Multi-party Computation (SMC) and homomorphic encryption. SMC prevents malicious server attacks but has high computational overhead, while homomorphic encryption requires a non-colluding external participant. On the other hand, differential privacy adds noise to intermediate parameters but is vulnerable to malicious clients. The proposed solution combines the strengths of differential privacy and weak encryption to mitigate these limitations.

Overall, the paper proposes a practical and effective solution for protecting data privacy in federated learning systems by combining differential privacy and weak encryption. This approach has the potential to be applied in various scenarios and can provide a higher level of security and privacy protection than existing methods.

5.6 Privacy Protection in Federated Learning

Federated learning is a distributed machine learning approach where multiple parties collaboratively train a model on their private data, without sharing their data with each other. However, as the parties involved in federated learning do not want to expose their private data, privacy preservation is a primary concern. Several techniques can be used to ensure privacy preservation in federated learning, including differential privacy, secure multi-party computation (SMPC), homomorphic encryption, and federated averaging.

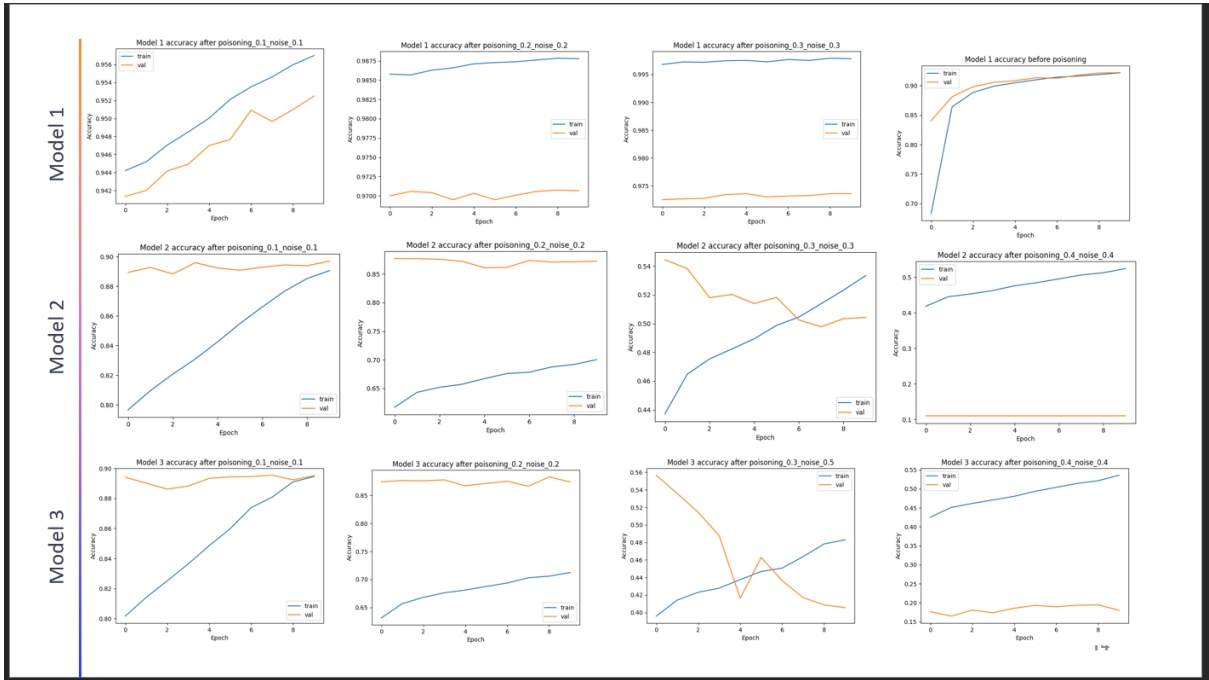
Differential privacy involves adding random noise to the training data to ensure that no individual's data can be inferred by an attacker. The amount of noise added is carefully calibrated to achieve a balance between privacy and model accuracy. SMPC allows multiple parties to securely compute a function over their private data without revealing their inputs to other parties. SMPC is based on cryptographic protocols that enable secure communication and computation between parties. Homomorphic encryption enables computations to be performed on encrypted data without decrypting it, thus preserving privacy. Parties can encrypt their data and send it to other parties for computation without exposing their private data.

Federated averaging involves aggregating locally trained models from multiple parties to create a global model while ensuring that individual party's data remains private. In this technique, each party trains a local model on their private data, and these models are then sent to a central server for aggregation. The central server aggregates the models and sends the updated model back to each party. The process repeats until a global model with high accuracy is achieved.

Overall, these techniques are used to ensure that the private data of each party is kept secure during the federated learning process, while still achieving a high level of accuracy in the final model.

6 Result Analysis

In our study, we conducted label flipping attacks on a selected dataset to analyze their impact on model accuracy in federated learning. Specifically, we applied two types of attacks - random and targeted - where we randomly changed the labels of a portion of the data for random flipping, and specifically changed the labels of data points belonging to a certain class for targeted flipping. To analyze the impact of these attacks, we developed a code that generated visualizations, allowing us to compare the effects of the two types of attacks and identify potential vulnerabilities in the dataset. Our results revealed that targeted label flipping had a greater impact on model accuracy compared to random label flipping, highlighting the importance of protecting against targeted attacks in federated learning.



The provided graphs depict the impact of targeted label flipping attacks on three different models. The first row represents the control model (Model 1) without any attacks. The second and third rows illustrate the effects of targeted label flipping attacks on two different classes (Class 1 to Class 7 for Model 2, and Class 3 to Class 8 for Model 3).

The last graph in the first row presents the accuracy of all three models before the attacks. The graphs in the second and third rows demonstrate the accuracy of the models after targeted label flipping attacks. The results indicate that as the noise or poison fraction increases beyond 0.3, the accuracy of the models decreases drastically, with the models unable to achieve a validation accuracy greater than 25%. These findings highlight the importance of protecting against targeted label flipping attacks in federated learning to ensure the model's accuracy and reliability.

6.1 Experimental Setup

Training Phase 1:

50 clients per round

100 rounds

Noise Multiplier - [0.0 - 1.1]

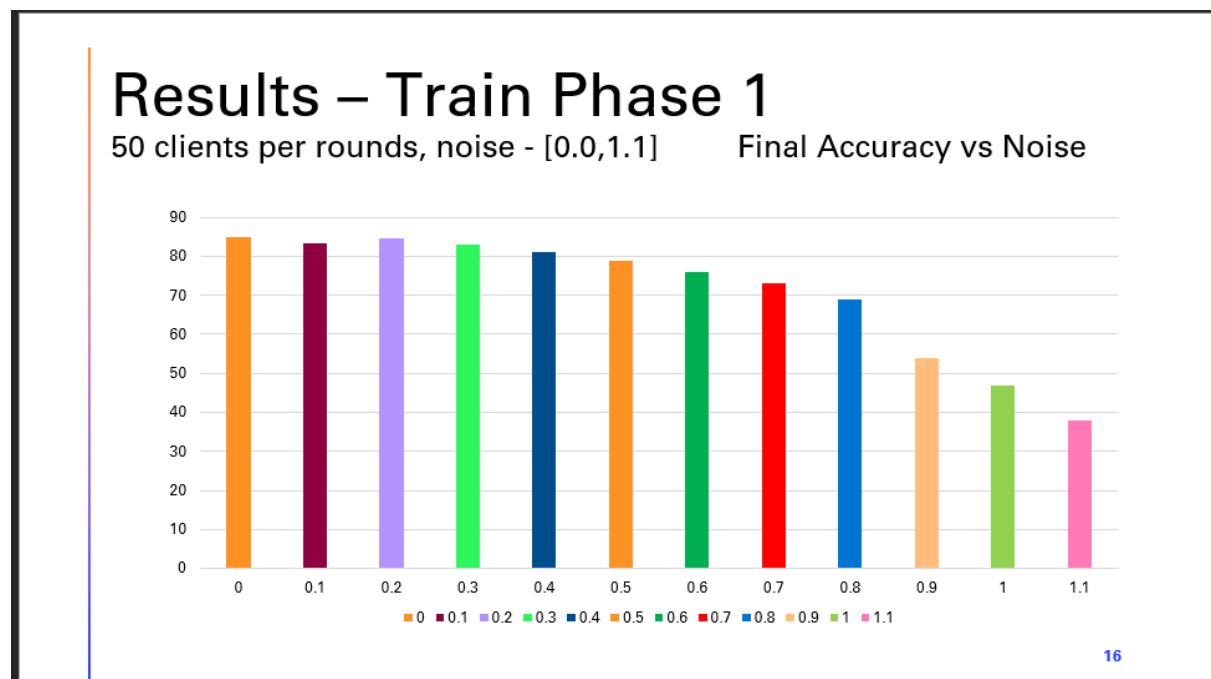
Training Phase 2:

120 clients per round

100 rounds

Noise Multipliers - [0.75 - 1.2]

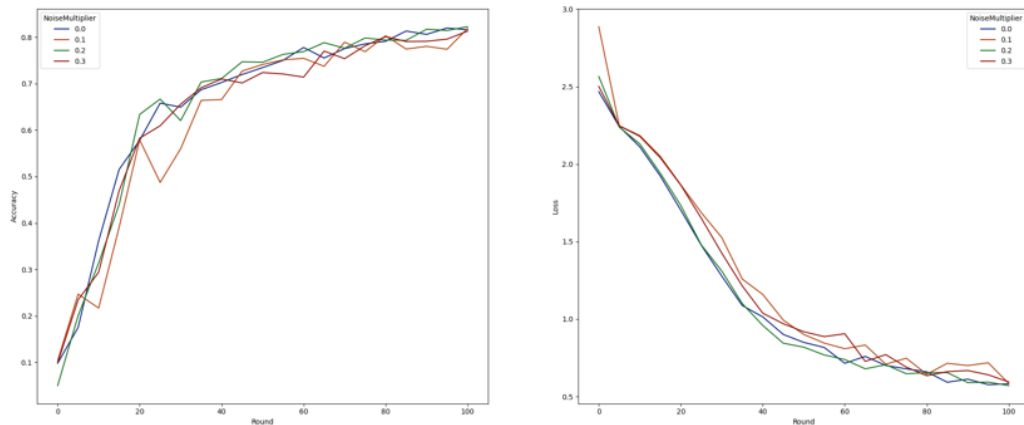
In this scenario, we trained multiple models with different levels of noise, using 50 clients per round. The noise was increased by adjusting the "noise_multiplier," which is the ratio of the noise standard deviation to the clipping norm. As the noise multiplier increased, the models became more private but less accurate. To determine the number of clients per round needed to achieve acceptable privacy, the team used dp_accounting functions and aimed for a delta value smaller than one over the number of records in the dataset (3383 training users), which was set at (2, 1e-5)-DP. The dp_accounting.calibrate_dp_mechanism was used to search for the optimal number of clients per round, and the RdpAccountant privacy accountant was based on Wang et al. (2018) and Mironov et al. (2019).



Here we can see the final accuracy achieved, during the training, by varying the noise_multiplier for different values.

Plots

Noise Multipliers [0-0.3]



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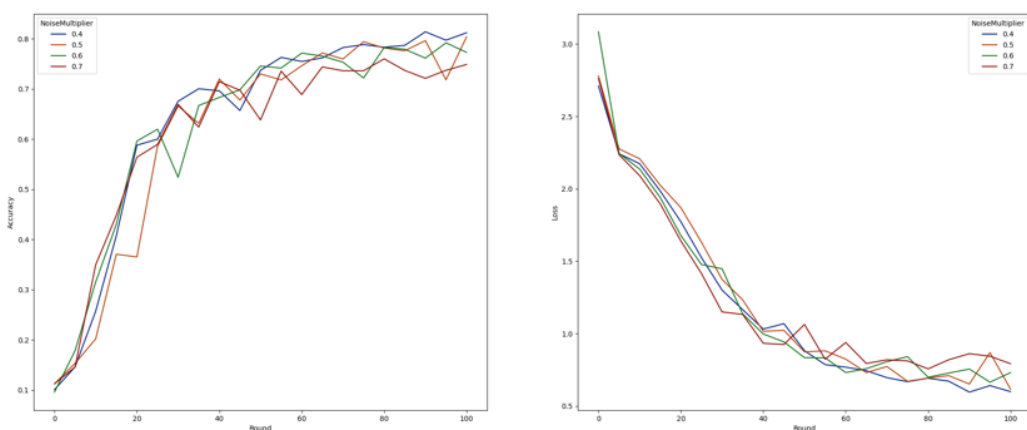
These plots are created with using 50 clients per round, and varying the noise fractions/multipliers in the range [0.0-1.1]

This plot shows the accuracy plots with noise values of 0.0, 0.1, 0.2, 0.3.

As we can see, the model gives quite similar performance with these noise values, and converges in the range of 83% - 85% for these noise values.

Plots

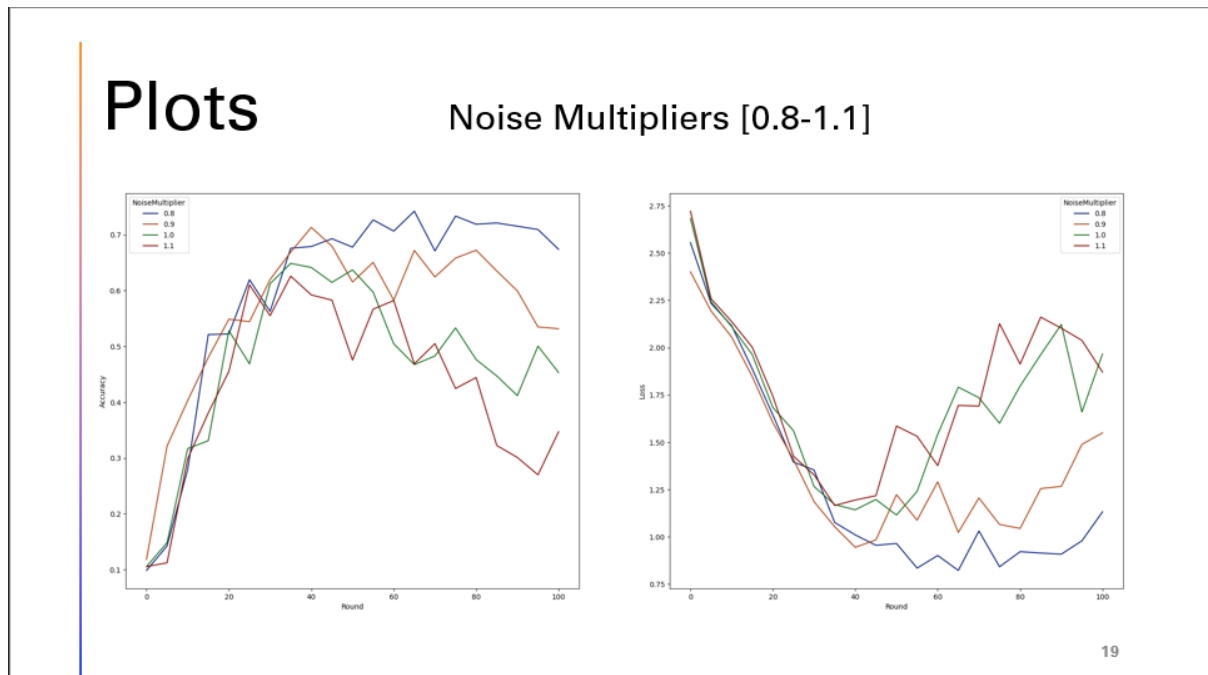
Noise Multipliers [0.4-0.7]



18

These plots are created with using 50 clients per round, and varying the noise fractions/multipliers in the range [0.0-1.1]

This plot shows the accuracy plots with noise values of 0.4, 0.5, 0.6, 0.7. Now we can see the accuracy of the model starts to drop for noise values above 0.5. It is also computationally expensive to train on noise values more than 0.5.



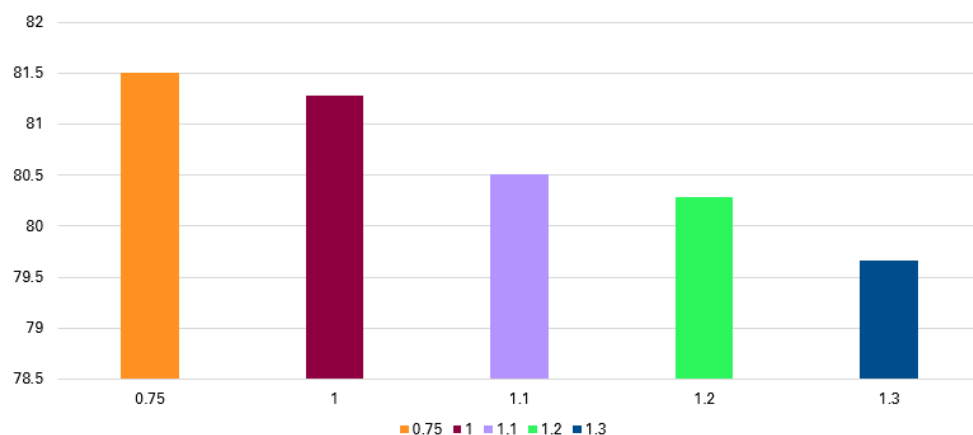
These plots are created with using 50 clients per round, and varying the noise fractions/multipliers in the range [0.0-1.1]

This plot shows the accuracy plots with noise values of 0.8, 0.9, 1.0, 1.1. Now we can see that the model accuracy suffers quite a lot for values greater than 0.75, and for values more than 0.9, the accuracies reach below 50%.

Results – Train Phase 2

120 Clients per round, noise - [0.75 - 1.3]

Final Accuracy vs Noise

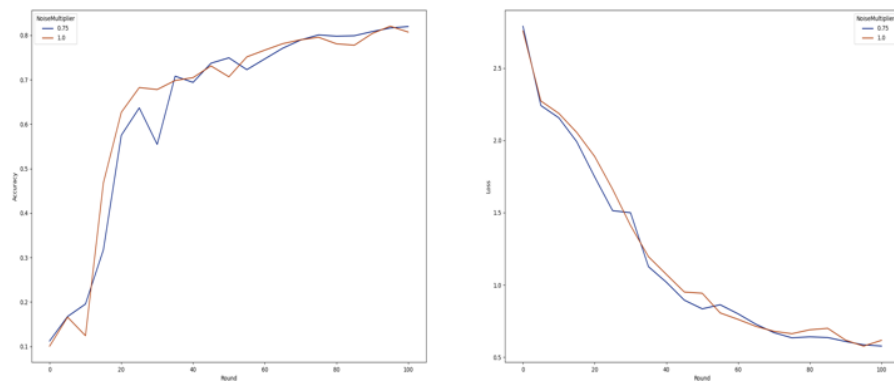


20

Here we can see the final accuracy achieved, during the training, by varying the noise_multiplier for different values, with Differential Privacy criteria being satisfied when using the noise multipliers as 1.2 or 1.3.

Plots

Noise Multipliers [0.75, 1.0]



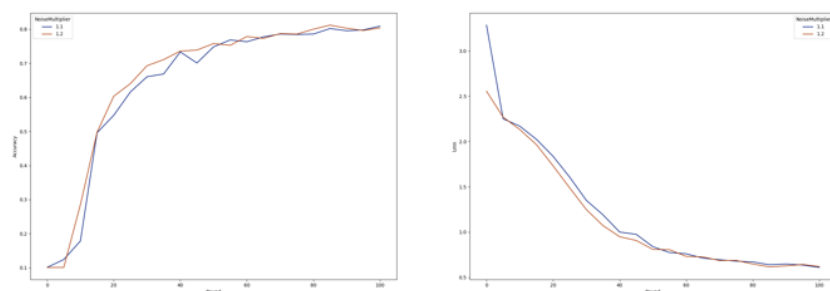
21

This plots are created with using 120 clients per round, and the noise fractions/multipliers used are 0.75 & 1.0

As we can see, this model performs better compared to the previous one, as we used 120 clients per round here compared to 50 in the previous one, and this converges in the range of 81% - 83% for these noise values.

Plots

Noise Multipliers [1.1, 1.2]



22

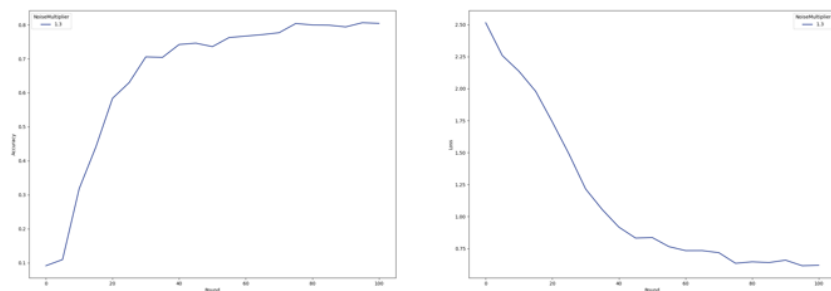
This plots are created with using 120 clients per round, and the noise fractions/multipliers used are 1.1 & 1.2

As we can see, this model performs better compared to the previous one, and the accuracy converges in the range of 80.6% +- 0.3% for these noise values.

This dataset has 3383 total training users, so delta would $1/3383 = (2, 1e-5)$ -DP. We searched over the number of clients per round and found that 120 clients per round and noise multiplier of 1.2 satisfy the privacy constraints.

Plots

Noise Multipliers [1.3]



23

This plots are created with using 120 clients per round, and the noise fractions/multipliers used is 1.3

The accuracy of this configuration converges in the range of 80.5 +- 0.5% for this noise values.

7 Conclusion

In summary, this project aimed to provide a comprehensive overview of privacy-preserving techniques in federated learning models. It began by introducing the fundamental concepts of federated learning, its advantages, disadvantages, and potential attacks. Then, the project implemented a label flipping attack and analyzed the impact of differential privacy techniques on the model's accuracy. The findings were presented through graphical analysis of model accuracy under various noise levels and targeted label flipping attacks.

In addition, the project identified potential areas for future research, including the use of homomorphic encryption and secure multi-party computation to enhance privacy and efficiency, as well as the incorporation of UMAP and weak encryption methods to improve privacy further.

Overall, this project provides valuable insights into the challenges and opportunities associated with preserving privacy in federated learning models and suggests promising directions for future research.

8 Future Work

Developing more sophisticated attack strategies: This could involve exploring novel approaches to model inversion attacks, membership inference attacks, backdoor attacks, or other privacy attacks. For example, developing new adversarial training techniques, or exploring the use of generative models to evade detection.

Designing more robust privacy-preserving techniques: This could include developing techniques such as secure multi-party computation, homomorphic encryption, differential privacy, or other techniques that can provide stronger guarantees of privacy in federated learning. For example, exploring the use of differential privacy for federated learning with a focus on reducing the impact on model performance.

Analyzing the impact of different federated learning settings on privacy and security: This could involve exploring the effects of different communication protocols, data distribution patterns, model architectures, or other factors on the vulnerability of federated learning systems to privacy attacks. For example, analyzing the effects of data heterogeneity or data imbalance on the privacy of federated learning models.

Developing techniques to optimize both privacy and model accuracy simultaneously: This could involve developing new optimization algorithms that can balance the trade-offs between privacy and model accuracy in federated learning. For example, exploring the use of privacy-aware regularization techniques or federated optimization algorithms that can optimize both objectives simultaneously.

Evaluating the real-world impact of privacy attacks on federated learning systems: This could involve collaborating with industry partners to test the effectiveness of different defense methods in practical settings, or conducting user studies to understand the impact of privacy attacks on user trust and behavior. For example, evaluating the impact of privacy attacks on medical diagnosis or financial prediction tasks in federated learning systems.

9 References

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10 Investigator Biographies

Dhyana Sai Reddy Dalli is a second-year master's student in computer science specializing in Data Science. He has experience working in a variety of fields, such as health care, risk management, software development, Full stack Developer.

Lakshmi Saieesha Bonthala holds a bachelor's degree in Electronics and Communication Engineering and is currently pursuing her master's in computer science at The University of Texas at Dallas, with a focus in Data Science. She started her career in NTT Data as a Software Engineer with 2+ years of experience. She has done Machine Learning as a graduate course and is looking forward to gaining expertise in the field of Data Science.

Neel Patel holds a bachelor's degree in information technology and is currently pursuing Master of Science in Computer Science at the University of Texas at Dallas with Information Assurance Track. He is NSA codebreaker 2022 solver. He possesses skills such as Binary Analysis, Software Reverse Engineering, Cryptography and programming skills in C++ and Python.

Nikhil Joshi is a MSCS Graduate Student at The University of Texas at Dallas. He is studying Cyber Security as his major. His coursework includes Network Security, Introduction to Cryptography, Malicious Code Analysis, Information Security, Data & Application Security, and Cloud Security. He regularly attends and practices CTF workshops and competitions.

Shrey Saini is a MSCS Graduate Student at The University of Texas at Dallas. He is studying Intelligence Systems As his major. He is well known for his knowledge in Python and Java Programming. He has immensely increased his knowledge space through his experience at Tata Consultancy Services where he worked as a systems Engineer. He is actively looking for Full Time Roles in the field of Intelligence Systems And Computer Science.

Srinivas Panchangam holds a bachelor's degree in electronics and communications Engineering and is pursuing his masters at The University of Texas at Dallas, graduating in Dec 2023 with a focus in Data Science. He started his career as a software developer in one of the biggest tech companies in India, TCS. Though being from a different bachelors background his interest towards data science never ended, so he is trying to explore the field and get expertise in that.